

3.1 Equipment

Item	Requirement	Why
VSX3000 sensor on a rigid stand	Stand mechanically separate from the robot base, on an isolated floor or table	Prevents robot motion from moving the sensor
6-axis robot	Repeatability $\leq$ ROBOT_REPEATABILITY_MM (ISO 9283); reports the actual, encoder-derived pose at POSE_LOG_RESOLUTION_MM, time-stamped against the frames; approach-from-below moves programmable; payload above the heaviest target plus adapter	Commanded, repeatable target poses; the read-back pose is the B-Z step truth
Quick-change target adapter	Dowel-pinned, re-mount repeatability $\leq$ ADAPTER_REMOUNT_REPEATABILITY_MM; every target carries the same dowel datum	Targets can be swapped without re-registering; the datum record places their features
Temperature loggers (sensor housing and air)	One sample every TEMPERATURE_LOG_INTERVAL_MIN	For drift attribution
Capture PC	VSX3000 SDK, LRVisionLibs MatCloud reader, robot interface that logs the actual pose read back from the robot	Synchronized capture and logging

3.2 Targets

ID	Target	Construction	Used in
T2	Noise and registration plate	Uniform matte, NOISE_PLATE_SIZE_MM, flatness PLATE_FLATNESS_MM, no pattern. Registration uses its plane (Section 4)	4, A, B-Z, sentinels
T3a	Raised square	Square of side EDGE_SQUARE_SIZE_MM with knife edges, on hidden posts at G above a back plate. Spacers set G to GAP_SMALL_MM or GAP_LARGE_MM.	B-HV, E
T3b	Square window	Front plate with a square window of the same size and knife edges; back plate at G behind it	B-HV, E
T4	Disk plate	Back-beveled disks on thin posts at G above a back plate: FEATURE_COUNT disks on the feature ladder, BLANK_SITES_PER_PLATE blank sites, and POST_SITES_PER_PLATE post-only control sites	C, D, E
T5	Cutout plate	Back-beveled holes in a front plate, with the back plate at G, removable for the open-background variant of Section 7: FEATURE_COUNT holes on the feature ladder and BLANK_SITES_PER_PLATE blank sites	C, D, E

Feature ladder. The sensor responds to the subtended size $D_{px} = D \cdot f_x / Z$. The robot sweeps each feature through the Z stations from the near limit of 400 mm to the far limit of 1600 mm, a factor of 4, so one feature covers two octaves of D_{px} . Diameters start at 3 pixels at the far station (FEATURE_MIN_PX_AT_Z_MAX) and grow by $2\sqrt{2}$ (FEATURE_LADDER_RATIO), so neighbors overlap by half an octave; the overlap is the scaling test of Sections 12 and 13. With 3 features per plate (FEATURE_COUNT) this gives, at $f_x \approx 688$ px (unconfirmed; superseded once the sensor's focal length, SENSOR_FX_PX, is known), 7.0, 19.7, and 55.8 mm, covering 3 to 96 px. The lower end sits below the minimum detectable size expected for the VSX3000: about 300,000 depth pixels share about 30,000 projected laser pencils, 10 pixels per pencil, and the path correlation needs at least 4 pencil detections, about 40 pixels of area or a diameter of 7 px; the matcher's support window (7 to 13 px in documented active-stereo processors) puts the plausible threshold at 10 to 15 px. Features on a plate are spaced at least 30 px apart at the far limit (FEATURE_ISOLATION_PX), so the sensor's spatial interpolation cannot couple neighboring features at any station. Plate extents follow from the features and the isolation: the plan tool's layout is the source of the as-built drawing, and each plate must fit the field of view at the near limit.

Blank and control sites. Each plate has 3 blank sites (BLANK_SITES_PER_PLATE), patches of back plate at least as large as the largest search window. These give the false-alarm rate in D at every station.

The disk plate also has 1 post-only site (POST_SITES_PER_PLATE), a post with no disk, which shows whether the support post itself is detected.

Datum and plate extents. Every target mounts on the same dowel datum, and the as-built record (Section 3.3) includes each feature's offset from that datum. Registration from the plane of T2 does not observe where a target sits sideways on the flange, so the once-per-mount check of Step 4.8 locates each mounted target in the camera frame: its front plane from the depth data (Z and tilt) and one feature edge in the left IR image (H and V). The back plates of T3a and T3b extend past the square by at least the boundary band plus the shadow at the 60 mm gap (GAP_LARGE_MM) and the worst ray angle of Section 3.3; the front plate of T3b and the plates of T4 and T5 extend past their features by at least the 30 px isolation at the far limit (FEATURE_ISOLATION_PX).

Surface finish. All front and back surfaces use one finish, for example bead-blasted aluminum or a matte coating, with mid-range IR reflectance. Record the finish and its measured reflectance at the projector wavelength in targets_asbuilt.csv. A reflectance difference between plates would bias the edge and area results.